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Provadis School of International Management and Technology

Exposé

Examining the Feasibility of Machine Learning-Based LogP Prediction on an ESP32 Microcontroller

A Proof-of-Concept Implementation and Performance Analysis

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Glossary

ESP32 Espressif Systems' low-cost, low-power system on a chip (SoC) with integrated Wi-Fi and Bluetooth capabilities. 1, 2

IoT Internet of Things. 1

logP Logarithm of the octanol-water partition coefficient. A measure of a compound's hydrophobicity, or how well it dissolves in fats versus water.. 1

TFLite TensorFlow Lite, a framework for deploying machine learning models on mobile and embedded devices. 1

ZINC20 ZINC Is Not Commercial (2020), a free database of commercially available compounds for virtual screening. 1

Exposé

Problem Statement

The accurate prediction of logP is crucial in various fields, including drug discovery and environmental chemistry. Traditional computational methods for logP prediction can be resource-intensive because they rely on storing large lists of fragments and their contributions to the overall logP value. This can make it challenging to deploy such models on resource-constrained devices like microcontrollers. Other, even more accurate methods, such as those based on quantum chemistry calculations, are even more computationally expensive and thus not suitable for deployment on microcontrollers. This project aims to explore the feasibility of implementing machine learning-based logP prediction on an ESP32 microcontroller, which is widely used in IoT applications.

Objectives

The primary objectives of this project are:

- To develop a machine learning model for logP prediction using a suitable dataset.
- To optimize the model for deployment on an ESP32 microcontroller, considering constraints such as memory and computational power.
- To evaluate the performance of the implemented model in terms of accuracy and efficiency.

Methodology

The project will follow these steps:

1. Data Collection: Gather a dataset of chemical compounds with known logP values. The ZINC20 database will be used as a primary source for this data.
2. Model Development: Train a machine learning model (e.g., Random Forest (RF), Support Vector Machine (SVM)) using the collected dataset and the TFLite framework for microcontroller deployment.

3. Model Optimization: Use techniques such as quantization and pruning to reduce the model size for deployment on the ESP32.
4. Implementation: Deploy the optimized model on the ESP32 microcontroller and develop an interface for inputting chemical compound data.
5. Evaluation: Assess the model's performance using metrics such as mean absolute error (MAE) and inference time on the ESP32.

Planned Structure

The paper will likely be structured as follows:

- Introduction
- Research Question and Objectives
- Literature Review
- Methodology
- Results and Discussion
- Limitations
- Conclusion and Future Work
- References
- AI Declaration
- Declaration of Authorship

Bibliography

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